# 1 What is SAP?

**SAP** stands for **Systems, Applications and Products in Data Processing**.

SAP is an **ERP (Enterprise Resource Planning) software** used by companies to manage all their business processes in one integrated system.

SAP software was developed by **SAP SE**.

Before SAP, companies used **different software for different departments**, which created many problems. SAP solved this by integrating everything into one system.

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# 2 What is ERP?

**ERP means Enterprise Resource Planning.**

ERP is a system used to manage all business departments together.

Departments in a company:

- Finance
- Sales
- Human Resources
- Inventory
- Production
- Purchasing

ERP integrates all these departments into **one system**.

Example ERP systems:

- **SAP S/4HANA**
- **Oracle ERP**
- **Microsoft Dynamics 365**

SAP is the **most widely used ERP system** in the world.

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# 3 Problems Before SAP (Real-Time Scenario)

Before SAP, companies used **separate systems**.

Example company departments:

## Finance Department

Used accounting software like **Tally**.

### HR Department

Employee details stored in **Microsoft Excel**.

### Sales Department

Used billing software or manual systems.

### Inventory Department

Used local database or separate software.

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### Real-Time Problem

Customer orders **100 laptops**.

Process before SAP:

1. Sales team writes order.
2. They inform warehouse manually.
3. Warehouse checks stock manually.
4. Finance creates invoice separately.
5. Management checks reports later.

Problems:

- Data mismatch
- Manual communication
- Time delay
- Errors in data

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## 4 How SAP Solved This Problem (Real-Time Example)

Customer orders **100 laptops**.

In SAP system:

1 Sales team creates **Sales Order**
2 SAP automatically checks **Inventory stock**
3 SAP creates **Delivery document**
4 Finance automatically gets **Invoice data**
5 Management sees **real-time reports**

Everything happens in **one system automatically**.

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## 5 Why Companies Use SAP

Companies use SAP because it helps to:

1. Integrate all departments
2. Reduce manual work
3. Provide real-time data
4. Improve business efficiency
5. Reduce errors
6. Improve reporting and analytics

Example companies using SAP:

- Tata Group
- Reliance Industries
- Infosys

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## 6 SAP Modules

SAP has different modules for different departments.

### SAP FICO

Finance and accounting.

### SAP MM

Material management and inventory.

### SAP SD

Sales and distribution.

### SAP HCM

Human resource management.

### SAP ABAP

Programming language used to develop SAP applications.

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## 7 Real-Time Example (Full Flow)

Customer orders a **mobile phone** from a company.

SAP process:

1. Sales creates **Sales Order (SD Module)**
2. Warehouse checks **Stock (MM Module)**
3. Delivery is created
4. Finance generates **Invoice (FICO Module)**

Payment is recorded

All data flows automatically inside SAP.

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## 8 SAP Database

SAP uses a powerful database called **SAP HANA**.

It stores and processes data very fast.

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### ⬙ What is SAP Fiori?

**SAP Fiori** is a **modern user interface (UI)** used to access SAP applications easily.

Developed by **SAP SE**

Simple:
**Fiori = Easy + User-friendly SAP screens (apps)**

### ⬙ Why SAP Fiori?

Old SAP (**SAP GUI**) problems:

- Complex screens
- Hard to use
- Not mobile friendly

Fiori solution:

- Simple design
- Mobile + Desktop support
- Easy navigation

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## 🔹 How Fiori Works?

1. User logs into Fiori Launchpad
2. Apps (tiles) are shown based on role
3. User clicks app
4. App connects to backend (**SAP S/4HANA**)
5. Data is displayed

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## 🔹 Key Concept

**Role-based apps**

Example:

- Manager → Reports
- Employee → Apply Leave
- Sales → Create Order

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## 🔹 Real-Time Example

**Leave Application**

Old way (SAP GUI):

- Enter T-code
- Complex screen

Fiori way:

- Click "Apply Leave" app
- Fill form
- Submit

Easy & fast

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## 🔹 Types of Fiori Apps

1. Transactional → Work (Create, Update)
2. Analytical → Reports, Charts
3. Fact Sheet → Display details

## ◈ Advantages

- User-friendly
- Faster
- Mobile support
- Less training required

## ◈ One Line

**SAP Fiori = Simple apps to use SAP easily**

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## What is SAP BTP?

**SAP BTP (Business Technology Platform)** is a **cloud platform** used to build, extend, and integrate SAP applications.

Developed by **SAP SE**

Simple:
**BTP = Platform to build apps outside SAP core**

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## ◈ Why SAP BTP?

Before BTP:

- Custom code directly in SAP system
- System becomes complex
- Upgrades become difficult

BTP Solution:

- Keep SAP core clean
- Build custom apps in cloud

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## ◈ Key Idea

**Clean Core Concept**

- Core SAP → Standard
- Custom Development → BTP

## What We Do in BTP?

Using BTP, we can:

- Build applications
- Create APIs
- Integrate systems
- Store and manage data
- Do analytics

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## How BTP Works?

1. User accesses application
2. App runs on BTP (cloud)
3. Connects to backend (**SAP S/4HANA**)
4. Data is fetched/updated

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## Technologies in BTP

- **CAP (Cloud Application Programming Model)**
- **Node.js / Java**
- OData Services
- SAP Fiori (UI)

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## Real-Time Example

### Scenario: Leave Approval App

Instead of developing inside SAP:

1. Build app in BTP
2. Connect to SAP system
3. Employee applies leave
4. Manager approves

SAP core remains clean

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## ◈ Advantages of BTP

- Keeps system clean
- Cloud-based (no local setup)
- Scalable
- Faster development
- Easy integration

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## ◈ One Line

**SAP BTP = Cloud platform to build and extend SAP applications**

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